

does not require you to explicitly specify the type of every variable you declare. Kotlin offers Data Classes, which automatically generate boilerplate code such as equals, hashCode, toString, getters/setters, and more. This significantly reduces the amount of code needed to define classes, as demonstrated by the following example:

```

• /* Java Code */
• class Book {
•     private String title;
•     private Author author;
•     public String getTitle()
•     {
•         return title;
•     }
•     public void setTitle(String title)
•     {
•         this.title = title;
•     }
•     public Author getAuthor()
•     {
•         return author;
•     }
•     public void setAuthor(Author author)
•     {
•         this.author = author;
•     }
• }
•
• /* Kotlin Code */
• data class Book(var title:String, var author:Author)

```

Kotlin is concise and reduces the amount of extra code written in other object-oriented programming languages. Additionally, Kotlin provides safety from NullPointerExceptions by supporting nullability as part of its system. Every variable in Kotlin is non-null by default, which means that assigning a null value to a non-null variable will result in a compile-time error. To assign null values to variables, they must be declared as nullable. The length() function is also disabled on nullable strings.

Kotlin is a programming language that possesses several features such as higher-order functions, lambda expressions, and operator overloading, that